Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT III : Probability Distributions

Syllabus: Binomial, Poisson and Normal distributions and their properties, Chebyshev's inequality, approximation of the binomial distribution to the normal distribution.

1. Introduction to Probability Distributions

Definition:

A probability distribution describes how the total probability 1 is spread over the values of a random variable.

- Discrete distributions: Bernoulli, Binomial, Poisson, Geometric, Hypergeometric.
- Continuous distributions: Uniform, Exponential, Normal, Gamma, t, χ^2, F.
- This unit: ***Binomial, Poisson, Normal***, Chebyshev's inequality, and normal approximation to binomial.

Distribution	Type	Parameters	Mean	Variance
Binomial	discrete	n, p	np	npq
Poisson	discrete	λ	λ	λ
Normal	continuous	μ, σ^2	μ	σ^2

Note:

Remember: q = 1 – p. Learn this table first – many answers start from it.

2. Bernoulli Trials and Binomial Distribution

Definition:

Bernoulli trial: a trial with only two outcomes - success (probability p) and failure (q = 1 - p).

- Bernoulli r.v.: P(X = 1) = p, P(X = 0) = q; E(X) = p, Var(X) = pq.

Binomial distribution

Definition:

X = number of successes in n independent Bernoulli trials, each with the same probability p of success. We write X ~ B(n, p).

Formula:

$P(X = x) = nCx \cdot p^x \cdot q^{n-x}$, x = 0, 1, 2, ..., n

Conditions

- Fixed number of trials n.
- Each trial has two outcomes (success / failure).
- Trials are independent; p is constant.
- Name: probabilities are the terms of the expansion $(q + p)^n$; so their sum = 1.

Properties of binomial distribution

Formula:

Mean = np, Variance = npq, SD = $\sqrt{(npq)}$

- Mean > Variance (since q < 1).
- Mode: the integer part of (n + 1)p. If (n + 1)p is an integer there are two modes: (n+1)p and (n+1)p - 1.
- Skewness $\gamma 1 = (q - p)/\sqrt{(npq)}$; kurtosis $\gamma 2 = (1 - 6pq)/(npq)$.
- If p = 0.5 symmetric; p < 0.5 positively skewed; p > 0.5 negatively skewed.

- Additive: $X1 \sim B(n1, p)$, $X2 \sim B(n2, p)$ independent $\Rightarrow X1 + X2 \sim B(n1 + n2, p)$.
- Recurrence: $P(x + 1) = [(n - x)/(x + 1)] \cdot (p/q) \cdot P(x)$.

Formula:

M.G.F.: $M(t) = (q + p\,e^{t})^{n}$

Derivation of mean

- $E(X) = \Sigma\, x\, nCx\, p^{x} q^{n-x} = np\, \Sigma\, (n-1)C(x-1)\, p^{x-1} q^{n-x} = np\,(q + p)^{n-1} = np.$

Example:

A fair coin is tossed 6 times. Find P(exactly 4 heads), P(at least 5 heads).

$n = 6, p = 1/2.\ P(X = 4) = 6C4\,(1/2)^{6} = 15/64 \approx$ **0.2344**

$P(X \geq 5) = 6C5/64 + 6C6/64 = (6 + 1)/64 =$ **7/64** $\approx$ **0.1094**

Example:

10 % of bolts are defective. A sample of 5 is taken. Find P(no defective), P(at most 1 defective).

$n = 5, p = 0.1, q = 0.9.\ P(0) = 0.9^{5} = 0.5905$

$P(1) = 5(0.1)(0.9)^{4} = 0.3281 \Rightarrow$ P(at most 1) $= 0.5905 + 0.3281 =$ **0.9186**

Example:

Mean of a binomial is 6 and variance is 4. Find n, p.

$np = 6, npq = 4 \Rightarrow q = 4/6 = 2/3, p = 1/3, n = 6/(1/3) =$ **18**

Example:

If 3 dice are thrown 100 times... for B(n = 10, p = 0.3): mean = 3, variance = 2.1, SD = 1.449.

3. Poisson Distribution

Definition:

Poisson distribution gives the probability of a number of events occurring in a fixed interval of time / space when events occur independently at a constant average rate λ. (Siméon Poisson, 1837)

Formula:

$P(X = x) = e^{-\lambda} \lambda^x / x!$, $x = 0, 1, 2, \ldots$ $X \sim P(\lambda)$

As a limit of binomial

- If $n \to \infty$, $p \to 0$ and $np = \lambda$ stays finite, then $B(n, p) \to \text{Poisson}(\lambda)$.
- Used for rare events: accidents, misprints per page, calls per minute, defects per metre, arrivals at a counter.
- Rule of thumb: $n \geq 20$ and $p \leq 0.05$ (or $n \geq 100$, $np \leq 10$).

Properties

Formula:

Mean = λ, Variance = λ, SD = $\sqrt{\lambda}$

- Mean = Variance is the characteristic property of the Poisson distribution.
- Mode: integer part of λ; if λ is an integer, two modes λ and λ - 1.
- Always positively skewed: $\gamma_1 = 1/\sqrt{\lambda}$, $\gamma_2 = 1/\lambda$.
- Additive: $X_1 \sim P(\lambda_1)$, $X_2 \sim P(\lambda_2)$ independent $\Rightarrow X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$.
- Recurrence: $P(x + 1) = [\lambda/(x + 1)]\, P(x)$, starting $P(0) = e^{-\lambda}$.

Formula:

M.G.F.: $M(t) = \exp[\lambda(e^t - 1)]$

Derivation of mean

- $E(X) = \Sigma\, x\, e^{-\lambda}\lambda^x/x! = \lambda\, e^{-\lambda}\, \Sigma\, \lambda^{x-1}/(x-1)! = \lambda\, e^{-\lambda}\, e^{\lambda} = \lambda$.

Example:

A call centre receives 3 calls per minute on average. Find P(exactly 2 calls) and P(more than 1) in a minute.

$\lambda = 3$, $e^{-3} = 0.0498$. $P(2) = 0.0498 \times 9/2 =$ ***0.2240***

$P(0) = 0.0498$, $P(1) = 0.1494 \Rightarrow P(X > 1) = 1 - 0.1992 =$ ***0.8008***

Example:

A book has 2 misprints per page on average. P(a page has no misprint)?

$\lambda = 2$: $P(0) = e^{-2} =$ ***0.1353***

Example:

1 % of items are defective. In a lot of 200 find P(at most 2 defective) using Poisson.

$\lambda = np = 200 \times 0.01 = 2$. $P(X \leq 2) = e^{-2}(1 + 2 + 2) = 5 \times 0.1353 =$ ***0.6767***

Example:

If $P(X = 1) = P(X = 2)$ for Poisson, find $P(X = 0)$.

$\lambda = \lambda^2/2 \Rightarrow \lambda = 2$. $P(0) = e^{-2} =$ ***0.1353***

Binomial	Poisson
Fixed n, two outcomes	Counts rare events, n large
Parameters n, p	One parameter λ
Mean np > variance npq	Mean = variance = λ
x = 0 to n	x = 0, 1, 2, ... (infinite)

4. Normal Distribution

Definition:

A continuous r.v. X follows the normal distribution with mean μ and standard deviation σ, written $X \sim N(\mu, \sigma^2)$, if its p.d.f. is

Formula:

$$f(x) = [1 / (\sigma\sqrt{(2\pi)})] \cdot \exp[-(x - \mu)^2 / (2\sigma^2)], -\infty < x < \infty$$

- Discovered by De Moivre; developed by Laplace and Gauss ("Gaussian distribution").
- Parameters: $-\infty < \mu < \infty$, $\sigma > 0$. Most important distribution in statistics (heights, marks, errors, ...).

Properties of the normal curve

- Bell shaped, symmetric about $x = \mu$.
- Mean = Median = Mode = μ; Variance = σ^2.
- Skewness = 0; kurtosis $\beta_2 = 3$ (mesokurtic).
- Maximum ordinate at $x = \mu$ equals $1/(\sigma\sqrt{(2\pi)})$.
- Points of inflexion at $x = \mu \pm \sigma$.
- x-axis is an asymptote; total area under curve = 1.
- Mean deviation = 0.7979σ ($\approx 4\sigma/5$); quartile deviation = 0.6745σ; Q.D. : M.D. : S.D. = 10 : 12 : 15.
- Odd order central moments vanish; $\mu_{2n} = 1 \cdot 3 \cdot 5 \ldots (2n - 1)\ \sigma^{2n}$.

Formula:

M.G.F.: $M(t) = \exp(\mu t + \sigma^2 t^2/2)$

- Linear combinations: $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$ independent $\Rightarrow aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$.

Area property (empirical rule)

Range	Area (probability)
$\mu \pm \sigma$	68.27 %

$\mu \pm 2\sigma$	95.45 %
$\mu \pm 3\sigma$	99.73 %

Standard normal distribution

Definition:

If $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$, with p.d.f. $\varphi(z) = (1/\sqrt{(2\pi)}) \exp(-z^2/2)$.

- Mean 0, variance 1. Table gives $\Phi(z) = P(Z \leq z)$.

Formula:

$P(Z \leq -z) = 1 - \Phi(z)$; $P(a < Z < b) = \Phi(b) - \Phi(a)$; $P(Z > z) = 1 - \Phi(z)$

z	Φ(z)
0.00	0.5000
1.00	0.8413
1.28	0.8997
1.645	0.9500
1.96	0.9750
2.00	0.9772
2.33	0.9901
2.58	0.9951
3.00	0.9987

Note:

Steps: 1) standardise $z = (x - \mu)/\sigma$ 2) sketch the curve 3) use symmetry and the table.

5. Worked Examples on Normal Distribution

Example:

$X \sim N(50, 100)$. Find $P(X > 65)$, $P(40 < X < 60)$.

$\sigma = 10$. $z = (65 - 50)/10 = 1.5 \Rightarrow P(Z > 1.5) = 1 - 0.9332 =$ **0.0668**

$P(40 < X < 60) = P(-1 < Z < 1) = 0.8413 - 0.1587 =$ **0.6826**

Example:

Marks are normal with mean 60, SD 12. What percentage of students score above 75? What is the minimum mark of the top 10 %?

$z = (75 - 60)/12 = 1.25 \Rightarrow P(Z > 1.25) = 1 - 0.8944 = 0.1056 \Rightarrow$ **10.56 %**

Top 10 % $\Rightarrow z = 1.28$. $x = 60 + 1.28 \times 12 =$ **75.4** marks.

Example:

Weights of 1000 students are N(65, 25) kg. How many weigh between 60 and 70 kg?

$\sigma = 5$, $z = \pm 1 \Rightarrow$ area $= 0.6826 \Rightarrow 1000 \times 0.6826 \approx$ **683** students.

Example:

For a normal variate, 7 % of items are below 35 and 89 % are below 63. Find μ and σ.

$P(X < 35) = 0.07 \Rightarrow z = -1.48$; $P(X < 63) = 0.89 \Rightarrow z = 1.23$

$(35 - \mu)/\sigma = -1.48$ and $(63 - \mu)/\sigma = 1.23$. Subtract: $28/\sigma = 2.71 \Rightarrow \sigma = 10.33$, $\mu = 35 + 1.48 \times 10.33 =$ **50.3**

Example:

A machine makes rods of mean length 10 cm and SD 0.05 cm. Rods outside 9.9 – 10.1 are rejected. Find percentage rejected.

$z = \pm 2 \Rightarrow$ accepted 0.9545 $\Rightarrow$ rejected $= 1 - 0.9545 =$ **4.55 %**

6. Chebyshev's Inequality

Definition:

For any random variable X with mean μ and finite variance σ^2, and any k > 0:

Formula:

$P(|X - \mu| \geq k\sigma) \leq 1/k^2$ equivalently $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$

- Another form: $P(|X - \mu| \geq c) \leq \sigma^2/c^2$ $(c > 0)$.
- It holds for ***every*** distribution, discrete or continuous, so the bound is conservative.

Proof (continuous case)

- $\sigma^2 = \int (x - \mu)^2 f(x)\, dx \geq \int$ over $|x - \mu| \geq k\sigma$ of $(x - \mu)^2 f(x)\, dx$.
- On this region $(x - \mu)^2 \geq k^2\sigma^2$, therefore $\sigma^2 \geq k^2\sigma^2 P(|X - \mu| \geq k\sigma)$.
- Divide by $k^2\sigma^2$: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$. ∎

k	Chebyshev: at least	Normal actual
2	75 %	95.45 %
3	88.9 %	99.73 %
4	93.75 %	99.99 %

Example:

A r.v. has mean 12 and variance 9. Find a lower bound for $P(6 < X < 18)$.

$\sigma = 3$, $|X - 12| < 6 = 2\sigma \Rightarrow k = 2 \Rightarrow P \geq 1 - 1/4 =$ ***0.75***

Example:

For $X \sim B(n = 100, p = 0.5)$: mean 50, $\sigma = 5$. Show $P(40 < X < 60) \geq 0.75$. $k = 2$ gives $P(|X - 50| < 10) \geq 3/4$, i.e. ***at least 0.75***. (Actual value $\approx$ 0.96, far larger.)

Example:

A factory makes items with mean weight 50 g and SD 2 g. Use Chebyshev to bound P(weight outside 44-56).

$|X - 50| \geq 6 = 3\sigma \Rightarrow P \leq 1/9 \approx$ ***0.111***

Note:

Chebyshev is used for the weak law of large numbers: $P(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \to 0$ as $n \to \infty$.

7. Normal Approximation to the Binomial

Definition:

(De Moivre - Laplace theorem) If $X \sim B(n, p)$, then for large n, $Z = (X - np)/\sqrt{(npq)}$ is approximately $N(0, 1)$.

- Thus X is approximately $N(np, npq)$.
- Good approximation when $np > 5$ and $nq > 5$ (or $np \geq 10$, $nq \geq 10$); best when $p \approx 0.5$.
- If p is very small ($np < 5$) use the Poisson approximation instead.

Continuity correction

- Binomial is discrete, normal is continuous: treat integer x as the interval $(x - 0.5, x + 0.5)$.

Binomial	Normal (continuity corrected)

$P(X = a)$	$P(a - 0.5 < X < a + 0.5)$
$P(X \le a)$	$P(X < a + 0.5)$
$P(X \ge a)$	$P(X > a - 0.5)$
$P(a \le X \le b)$	$P(a - 0.5 < X < b + 0.5)$

Example:

A fair coin is tossed 100 times. Find $P(X \ge 60)$ and $P(45 \le X \le 55)$.

$\mu = np = 50$, $\sigma = \sqrt{(100 \times 0.5 \times 0.5)} = 5$.

$P(X \ge 60) = P(X > 59.5)$: $z = (59.5 - 50)/5 = 1.9 \Rightarrow 1 - 0.9713 =$ ***0.0287***

$P(45 \le X \le 55) = P(44.5 < X < 55.5)$: $z = \pm 1.1 \Rightarrow 0.8643 - 0.1357 =$ ***0.7286***

Example:

A die is thrown 180 times. P(number of sixes ≤ 25)?

$n = 180$, $p = 1/6 \Rightarrow \mu = 30$, $\sigma = \sqrt{(180 \times 1/6 \times 5/6)} = \sqrt{25} = 5$.

$P(X \le 25) = P(X < 25.5)$: $z = (25.5 - 30)/5 = -0.9 \Rightarrow$ ***0.1841***

Example:

400 students; each is absent with probability 0.1. P(at most 30 absent)?

$\mu = 40$, $\sigma = \sqrt{(400 \times 0.1 \times 0.9)} = 6$. $P(X < 30.5)$: $z = -9.5/6 = -1.583 \Rightarrow \approx$ ***0.0567***

8. Practice Problems with Answers

- 1) A fair die is thrown 4 times. P(exactly two sixes)? Ans: $4C2\ (1/6)^2(5/6)^2 = 150/1296 = 0.1157$.
- 2) A binomial has mean 4 and variance 3. Find n, p. Ans: $q = 3/4$, $p = 1/4$, $n = 16$.

- 3) Poisson with mean 4: find $P(X \leq 2)$. Ans: $e^{-4}(1 + 4 + 8) = 13 \times 0.0183 = 0.2381$.
- 4) $X \sim N(100, 25)$. Find $P(95 < X < 110)$. Ans: z from -1 to 2: 0.9772 – 0.1587 = 0.8185.
- 5) Show that the normal curve has points of inflexion at $\mu \pm \sigma$. Hint: set $f''(x) = 0$.
- 6) Prove mean of Poisson = variance = λ using the m.g.f. Hint: $M'(0)$, $M''(0)$.

Note:
Quick decision: count with fixed n → binomial; rare events per interval → Poisson; measurements / large n → normal.

9. More on Binomial Distribution

Variance of binomial (derivation)

- $E[X(X - 1)] = \Sigma\, x(x - 1)\, nCx\, p^x q^{n-x} = n(n - 1)p^2 \Sigma\, (n-2)C(x-2)\, p^{x-2} q^{n-x} = n(n - 1)p^2$.
- $Var(X) = E[X(X - 1)] + E(X) - [E(X)]^2 = n(n - 1)p^2 + np - n^2p^2 = np - np^2 = \mathbf{npq}$.

Fitting a binomial distribution

- Step 1: find p from the data (mean = np). Step 2: compute P(x). Step 3: expected frequency = N × P(x).

Example:
4 coins are tossed 160 times. Find expected frequencies of 0, 1, 2, 3, 4 heads.

Heads x	0	1	2	3	4
P(x)	1/16	4/16	6/16	4/16	1/16
Expected f	10	40	60	40	10

Example:

Ten fair coins tossed. Find $P(X = 5)$ and $P(X \leq 2)$.

$P(5) = 10C5/1024 = 252/1024 = \mathbf{0.2461}$

$P(X \leq 2) = (1 + 10 + 45)/1024 = 56/1024 = \mathbf{0.0547}$

Example:

The chance a seed germinates is 0.25. 8 seeds are sown. Find P(at least one germinates).

$P = 1 - 0.75^8 = 1 - 0.1001 = \mathbf{0.8999}$

Example:

Find the mode of $B(n = 10, p = 0.4)$.

$(n + 1)p = 11 \times 0.4 = 4.4 \Rightarrow$ mode $= \mathbf{4}$

Example:

A multiple choice test has 5 questions, each with 4 options. A student guesses. P(at least 4 correct)?

$p = 1/4$. $P(4) = 5\,(1/4)^4(3/4) = 15/1024$; $P(5) = 1/1024$.

$P(X \geq 4) = 16/1024 = \mathbf{0.0156}$

Example:

In a binomial with $n = 5$, find the variance of the number of heads when 5 coins are tossed.

$npq = 5 \times 0.5 \times 0.5 = \mathbf{1.25}$

Note:

Maximum variance of $B(n, p)$ for fixed n occurs at $p = 1/2$ (npq is largest when $pq = 1/4$).

10. More on Poisson Distribution

Variance of Poisson (derivation)

- $E[X(X-1)] = \Sigma\, x(x-1)\, e^{-\lambda}\lambda^x/x! = \lambda^2 e^{-\lambda} \Sigma\, \lambda^{x-2}/(x-2)! = \lambda^2$.
- $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Fitting a Poisson distribution

- Mean of data $\bar{x} = \Sigma fx/\Sigma f = \lambda$. Expected frequency $= N \cdot e^{-\lambda}\lambda^x/x!$, use recurrence $P(x+1) = \lambda P(x)/(x+1)$.

Example:

Accidents occur at an average of 1.5 per day. Find P(0), P(1), P(2) and P(at least 3 accidents).

x	0	1	2
P(x)	0.2231	0.3347	0.2510

- $P(X \le 2) = 0.2231 + 0.3347 + 0.2510 = 0.8088 \Rightarrow P(X \ge 3) = 1 - 0.8088 =$ **0.1912**

Example:

A binomial with n = 100, p = 0.02: use Poisson to find P(X = 3).

$\lambda = np = 2$. $P(3) = e^{-2} \times 2^3/3! = 0.1353 \times 1.3333 =$ **0.1804**

Example:

2 % of fuses are defective. In a box of 100 find P(at least one defective).

$\lambda = 2 \Rightarrow P(X \ge 1) = 1 - e^{-2} = 1 - 0.1353 =$ **0.8647**

Example:

A Poisson variable has $P(X = 0) = 0.3679$. Find its mean and $P(X = 2)$.

$e^{-\lambda} = 0.3679 \Rightarrow \lambda = 1$. $P(2) = e^{-1}/2 =$ **0.1839**

Example:

Number of emails per hour is Poisson with mean 4. Find P(at most 1 email in a half hour).

Half hour: $\lambda = 2$. $P(X \le 1) = e^{-2}(1 + 2) = 3 \times 0.1353 =$ **0.4060**

Note:

If the average rate is λ per unit time, then in t units the mean becomes λt.

II. More on Normal Distribution

Total area under the curve = 1 (proof sketch)

- Let $I = \int \exp(-z^2/2)\, dz$. Then $I^2 = \int\int \exp(-(x^2+y^2)/2)\, dx\, dy = \int$ from 0 to $2\pi \int$ from 0 to $\infty \exp(-r^2/2)\, r\, dr\, d\theta = 2\pi$.
- So $I = \sqrt{(2\pi)}$ and $\int \varphi(z)\, dz = 1$.

Mean and variance of $N(\mu, \sigma^2)$

- Put $z = (x - \mu)/\sigma$: $E(X) = \mu + \sigma E(Z)$, and $E(Z) = \int z\, \varphi(z)\, dz = 0$ (odd function) $\Rightarrow E(X) = \mu$.
- $Var(Z) = \int z^2\, \varphi(z)\, dz = 1$ (integration by parts) $\Rightarrow Var(X) = \sigma^2$.

Using the table – special areas

Required	Formula
$P(Z \le a)$, $a > 0$	$\Phi(a)$ directly
$P(Z \ge a)$, $a > 0$	$1 - \Phi(a)$
$P(-a \le Z \le a)$	$2\Phi(a) - 1$
$P(Z \le -a)$	$1 - \Phi(a)$

$P(a \le Z \le b)$	$\Phi(b) - \Phi(a)$

Example:

$X \sim N(30, 16)$. Find $P(X < 24)$ and $P(26 < X < 38)$.

$\sigma = 4$. $z = (24 - 30)/4 = -1.5 \Rightarrow P =$ **0.0668**

$P(26 < X < 38) = P(-1 < Z < 2) = 0.9772 - 0.1587 =$ **0.8185**

Example:

$X \sim N(70, 64)$. Find k with $P(X > k) = 0.05$.

$z = 1.645 \Rightarrow k = 70 + 1.645 \times 8 =$ **83.16**

Example:

Scores are $N(80, 100)$. Find percentage between 70 and 90, and above 100.

Between: $z = \pm 1 \Rightarrow$ **68.26 %**. Above 100: $z = 2 \Rightarrow 1 - 0.9772 =$ **2.28 %**

Example:

$X \sim N(10, 4)$ and $Y \sim N(20, 9)$ independent. Find the distribution of $X + Y$ and $X - Y$, and $P(X + Y > 35)$.

$X + Y \sim N(30, 13)$; $X - Y \sim N(-10, 13)$ (variances always add).

$z = (35 - 30)/\sqrt{13} = 5/3.606 = 1.39 \Rightarrow P = 1 - 0.9177 =$ **0.0823**

Example:

Heights of 500 men are $N(168, 36)$ cm. How many are taller than 180 cm?

$z = 12/6 = 2 \Rightarrow 0.0228 \times 500 \approx$ **11** men.

Example:

The life of a bulb is N(1200, 3600) hours. Find the life L such that 90 % of bulbs last more than L.

$P(X > L) = 0.9 \Rightarrow z = -1.28 \Rightarrow L = 1200 - 1.28 \times 60 =$ **1123.2 hours**

12. More on Chebyshev and Approximations

Example:

Mean 100, SD 10. At least what fraction lies between 80 and 120?

$k = 2 \Rightarrow$ at least $1 - 1/4 =$ **75 %**

Example:

How many standard deviations around the mean guarantee at least 96 % of the data?

$1 - 1/k^2 = 0.96 \Rightarrow k^2 = 25 \Rightarrow$ **$k = 5$**

Example:

The number of items sold has mean 80, variance 25. Lower bound for $P(70 < X < 90)$?

$|X - 80| < 10 = 2\sigma \Rightarrow P \geq$ **0.75**

Example:

Using Chebyshev, estimate the number of tosses so that $P(|\hat{p} - 0.5| < 0.1) \geq 0.95$.

$\sigma^2 = 0.25/n$; need $0.25/(n \times 0.01) \leq 0.05 \Rightarrow n \geq$ **500**

Normal approximation – more examples

Example:

A fair coin is tossed 200 times. Find $P(X \geq 110)$.

$\mu = 100$, $\sigma = \sqrt{50} = 7.07$. $P(X > 109.5)$: $z = 9.5/7.07 = 1.34 \Rightarrow 1 - 0.9099 = \approx$ **0.090**

Example:

$X \sim B(20, 0.5)$. Approximate $P(X = 10)$.

$\mu = 10$, $\sigma = 2.236$. $P(9.5 < X < 10.5) = P(-0.22 < Z < 0.22) = 2(0.5871) - 1 =$ **0.174** (exact 0.1762)

Example:

Binomial $n = 500$, $p = 0.004$ – which approximation? $np = 2 < 5 \Rightarrow$ **Poisson with $\lambda = 2$**, not normal.

Distribution	Approximated by	When
Binomial	Poisson(np)	n large, p small
Binomial	Normal(np, npq)	n large, p not small
Poisson	Normal(λ, λ)	λ large (≥ 10)

13. Practice Problems with Answers

- 1) A coin is tossed 5 times. P(exactly 3 heads)? Ans: 10/32 = 0.3125.
- 2) A binomial has $n = 6$ and $9P(X = 4) = P(X = 2)$. Find p. Ans: $9 \times 15\, p^4 q^2 = 15\, p^2 q^4 \Rightarrow 9p^2 = q^2 \Rightarrow 3p = q \Rightarrow p = 1/4$.
- 3) Poisson mean 3: $P(X > 4)$? Ans: $1 - e^{-3}(1 + 3 + 4.5 + 4.5 + 3.375) = 1 - 0.8153 = 0.1847$.
- 4) $X \sim N(50, 16)$. Find $P(X > 58)$. Ans: $z = 2 \Rightarrow 0.0228$.
- 5) Prove that $E(X) = \lambda$ for Poisson using the m.g.f. Hint: $M'(t) = \lambda e^t M(t)$.

- 6) Show $\beta_2 = 3$ for the normal distribution. Hint: $\mu_4 = 3\sigma^4$.
- 7) A r.v. has mean 5 and variance 4. Bound $P(|X - 5| \geq 6)$. Ans: $\leq 4/36 = 1/9$.
- 8) 30 % of items are defective; sample of 100: approximate $P(X \leq 35)$. Ans: $\mu = 30$, $\sigma = 4.58$, $z = 5.5/4.58 = 1.2 \Rightarrow 0.8849$.

Quick Revision – Unit III

- Binomial: $nCx\ p^x q^{n-x}$; mean np, var npq.
- Poisson: $e^{-\lambda}\lambda^x/x!$; mean = var = λ.
- Normal: $Z = (X - \mu)/\sigma$; 68 – 95 – 99.7 rule.
- Chebyshev: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$.
- Binomial → Normal: $N(np, npq)$ with continuity correction ± 0.5.